University of Toronto Scarborough

CSC/MAT A67 **Term Test 1** **12 February 2022**

NAME: ______________________________________
(please print; circle your last name)

SIGNATURE: ____________________

Circle your tutorial section:

TUT01 MO 9-10 MW 140 Ali	TUT02 MO 9-10 HL B110 Vishal	TUT03 TU 19-20 AA 204 Ananya	TUT04 TH 14-15 IC 120 Aamir	TUT05 FR 9-10 HL B110 Mohannad	TUT06 FR 14-15 SW 143 Irene

Do not begin until you are told to do so.
In the meantime, complete the above and read the rest of this cover page.

Aids allowed: None.

Duration: 90 minutes.

There are 4 pages and each is numbered at the bottom. Make sure you have all of them.

Write legibly on the front of each page. **What you write on backs of pages will not be graded.**

Question	Your mark	Out of
1.		10
2.		10
3.		10
Total		30

1. For $j = 0, 1, 2$, let $P_j(n)$ be the statement "there exists an integer k such that $n = 3k + j$".

   The following fact can be used without proof for this question.

   Fact: If n is an integer, then exactly one of $P_0(n), P_1(n), P_2(n)$ is true.

   Let n be an arbitrary integer.
   Use a proof by contraposition to prove that if $P_0(n^2)$ is true, then so is $P_0(n)$.

2. Use a proof by contradiction to prove that $\sqrt{3}$ is irrational.

   You may use the result from question 1, even if you did not complete it.

3. Use principle of mathematical induction to prove that $\sum_{i=0}^{n} \frac{2}{3^i} < 3$ for all nonnegative integers n.
Hint: You need to prove a stronger statement.